Smart Parking System – Concept & System Design

Nov 13, 2026 · @JOne Nakasamai

Part A – Concept

1. Executive summary

A plate-first, cashless parking platform for Fiji's private car parks and municipalities, built software-first. Phase 1 proves the concept with phone-based number-plate recognition; an RFID windscreen sticker follows as a phase 2 upgrade.

- **What it does:** drivers pay for parking by plate with M-PAiSA, MyCash or card. They get reminders before expiry and can extend from their phone. Officers check any car by pointing a smartphone at its plate. Notices are issued digitally, using Book Now or Delayed Booking rules each car park owner sets.
- **Four products, one backend:** Driver app, Officer app, Operator admin (for the car park owner) and Platform admin (for us), all on Google Cloud / Firebase.
- **Hardware in phase 1:** none beyond the officer's phone. Phase 2 adds RFID stickers and readers, and phase 3 adds gate cameras and readers for gated car parks.
- **Timeline:** a pilot-ready MVP in about 16 weeks, an 8-week plate-only pilot, then commercial launch. RFID hardware runs from about week 28 (Part D).
- **Differentiators:** an enforcement engine the owner configures (grace periods, delayed booking, penalties); one driver account that works across every participating car park in Fiji; works offline; and an upgrade path to RFID speed and tamper-proof vehicle identity.

2. Market & positioning

- **Suva has an incumbent.** In January 2026 Suva City Council launched **Bula Parki**, built by D2UNET. It offers cashless payments, real-time monitoring and expiry alerts. It's rolling out in council-owned car parks first, and to inner-city streets only after law amendments. Broken meters with no spare parts drove the change.
- **Other towns are moving too.** Government approved traffic and parking improvements for Labasa, Nadi, Tavua and Savusavu in July 2026, and Savusavu has digital display meters.

Go-to-market order

1. **Private car parks** (fastest decisions): malls, hotels, hospitals, universities (USP, FNU), office blocks, airport-area operators in Nadi.
2. **Other municipalities:** Lautoka, Nadi, Nausori, Nasinu, Lami, Labasa, Ba, Sigatoka, Savusavu.
3. **Suva City Council:** offer RFID enforcement and multi-operator integration *alongside* Bula Parki, not as a replacement.

2.1 Monetisation: charge both, in a balanced way

Recommendation: a hybrid model. Each operator pays a monthly plan fee that covers our fixed cost of serving them. A small fee on every paid transaction scales with the value we process. The operator chooses whether that fee is passed on to the driver or absorbed.

Revenue stream	Who pays	How	Placeholder price (FJD, to test in the pilot)
Plan subscription	Operator	Monthly, per site, by plan (Section 2.2)	Starter FJ$0–99 · Professional FJ$299 · Enterprise/Municipal custom
Transaction fee	Operator (from settlement) **or** driver (convenience fee)	Per paid session or exit	Starter 8% · Professional 5% · Enterprise 2–3%, or a fixed FJ$0.20–0.50 driver convenience fee
Notice processing fee	Operator	Share of penalties collected through the platform	10–15% of collected penalties, or a fixed fee per notice
Gate lane service	Operator	Installation (one-off) + monthly per-lane support and software fee	Install at cost + margin; FJ$150 per lane per month
Add-on modules	Operator	Monthly per module (Section 2.2)	FJ$50–200 per module
Fleet accounts	Business customer	Per registered vehicle per month (portal, invoicing, reports)	FJ$5–10 per vehicle per month
RFID stickers (phase 2)	Driver or fleet	Retail margin; replacements	Price to set after supplier quotes

Revenue stream	Who pays	How	Placeholder price (FJD, to test in the pilot)
SMS beyond plan quota	Operator	Bundles	Cost + margin
Advertising (optional)	Advertisers	Exit screens and the Driver app's session screen, sold as digital signage inventory	Revenue share with the operator

Rules that keep it fair and simple

- Councils usually buy through tenders, so offer them an annual licence with a low transaction share rather than high per-transaction fees.
- When we're merchant of record, our fees are deducted automatically from each daily settlement. Operators with their own merchant accounts get a monthly invoice.
- The driver convenience fee is shown before payment and never applies to paying a notice.
- Parking credit top-ups carry no driver fee, to encourage bigger top-ups that cut mobile money costs.
- All prices above are placeholders to validate with pilot operators; they are not market research.

2.2 Plans & feature modules

The platform is **modular**: every major feature is a module the App Owner (Platform admin) switches on or off per operator, based on the plan they subscribe to and any add-ons they buy.

Module	Starter	Professional	Enterprise / Municipal
Core: driver payments, sessions, reminders, Officer app with plate recognition, Book Now, basic dashboard	Included	Included	Included
Sites / bays / officers included	1 / 100 / 2	3 / 500 / 10	Unlimited
Delayed Booking & advanced enforcement policies	Add-on	Included	Included

Module	Starter	Professional	Enterprise / Municipal
Basic permits & exemptions	Included	Included	Included
Advanced passes (scheduled, visitor, validation) & tenant sub-accounts	—	Included	Included
Gated car park (boom gates, per lane)	—	Add-on per lane	Add-on per lane
Fleet accounts & deliveries	—	Add-on	Included
Parking credit wallet	Included once launched	Included	Included
Advanced analytics & scheduled reports	—	Included	Included
USSD / SMS channel	Add-on	Add-on	Included
Paper notices (Bluetooth printer)	Add-on	Add-on	Included
RFID stickers & readers (phase 2)	—	Add-on	Add-on
Partner API & webhooks	—	Add-on	Included
White-label branding (own app name/logo)	—	—	Add-on
SLA & priority support	Standard	Business hours	24/7 P1 response

How the toggles work (technical)

- **Entitlements in Firestore:** `plans` (modules + limits), `subscriptions` (operator, plan, add-ons, status, billing period), and a computed `entitlements` record per operator.
- **Enforced on the server:** every Cloud Function and security rule checks entitlements, so a hidden button is never the only protection. Apps read entitlements to show or hide screens and menus.
- **Limits enforced softly:** warnings at 80%, then a block on creating new sites, officers or lanes over the limit. Existing activity never stops mid-day.
- **Feature flags vs entitlements:** Remote Config controls technical rollout (e.g. turning on a new feature for 10% of users). Entitlements control what a customer has paid for. The two are kept separate.

- **Plan changes:** upgrades take effect immediately (prorated). Downgrades take effect at the next billing period. Trials last 30 days on Professional.
- **Non-payment:** a 14-day grace period, then read-only admin. **Safety rule: gates are never locked shut** for an unpaid subscription; exits always open.
- Every toggle change is in the audit log, with who made it and why.

3. Users & roles

Role	Who	Main device	Core need
Driver	Car owner or regular driver; business fleet manager	Smartphone app, web, or USSD/SMS on any phone	Pay quickly, never get an unfair fine
Officer	Council parking warden or private car park attendant	Android phone running the Officer app	Check a car in seconds; issue notices that stand up to a dispute
Operator admin	Council parking manager or car park owner/manager	Web (desktop or tablet)	Set rates and rules, see revenue and compliance, handle disputes
Operator finance	Council or company accounts staff	Web	Settlement reports, reconciliation, refunds
Sales agent	Mobile money agent, reseller or council counter	Agent app (a mode of the Officer app)	Sign up drivers who don't use smartphones; later, sell and fit stickers
Platform admin	Us: operations, support, finance, security	Web	Onboard operators, review identity checks, reconcile payments, support users

4. End-to-end user journeys

4.1 Driver: first time parking (target under 2 minutes)

1. Arrives at the car park and sees signage with the zone code and a QR code (e.g. *"Pay by phone – Zone SV-104"*).

2. Scans the QR, which opens the web app pre-filled with that zone. They can also install the app from Google Play or the App Store.
3. Enters their phone number and the SMS one-time code.
4. Adds the vehicle: plate, make and colour.
5. Picks a duration. The price updates live (e.g. *2 h = FJ$2.00*).
6. Pays with M-PAiSA, MyCash or card, and gets a confirmation screen with a countdown and a signed QR pass.
7. Reminders arrive 15 and 5 minutes before expiry, with an **Extend** button.
8. The identity check (licence + live selfie) is prompted **after** the first payment, and required before the second session or before any notice can be disputed. The first park stays fast.

4.2 Driver: returning, extending and leaving

1. Opens the app. The last vehicle and a nearby zone are pre-selected by GPS, so it's **one tap to pay**.
2. Extends from the reminder notification, without opening the app.
3. Taps **End parking early** (if the operator refunds unused time).

4.3 Driver: delayed booking and notices

1. Officer starts a Delayed Booking. The driver gets push + SMS: *"Parking expired at Zone SV-104. Top up within 10 min to avoid a FJ$20 fine."*
2. One tap opens checkout with the overdue minutes already included. Paid → *"Resolved – no fine."*
3. If a notice is issued, the driver sees officer photos, time, location and offence code, then chooses **Pay** or **Dispute**.
4. A dispute takes a reason, a photo or receipt upload, and gives a decision deadline. The outcome arrives by push, SMS and email.

4.4 Driver without a smartphone

1. Registers by USSD (`*xxx#`) or at an agent who captures their licence and photo.
2. Pays by USSD menu or SMS (`PARK SV104 2 LT1234`), charged to M-PAiSA or MyCash.
3. Gets SMS reminders and notices. Can dispute through an agent or the support line.

4.5 Officer: a shift

1. Logs in on a registered phone with staff ID + PIN or biometric. The zone snapshot downloads.

2. Opens *Patrol*. The beat map shows zones and the cars due to expire.
3. Points the camera at a plate. Plate recognition reads it, the officer taps to confirm, and the status card appears (green, amber, red, blue or grey).
4. On red: **Book Now** (2+ photos, offence code, submit) or **Delayed Booking** (starts a timer and goes on the revisit list).
5. The revisit list buzzes when a delayed booking expires. The officer goes back, takes a fresh photo and confirms, or it auto-resolves if the driver paid.
6. Ends the shift and sees a summary: cars checked, notices, delayed bookings resolved, distance walked.

4.6 Operator: setting up a car park (target under 1 day)

1. Platform admin creates the operator account and invites the operator admin.
2. The operator draws zones on a map, sets bays and hours, and builds rate tables.
3. Sets the enforcement policy (Section 10) and penalty codes from their by-laws or terms.
4. Invites officers and registers their phones.
5. Orders the signage kit (zone codes + QR) and publishes the car park. It now appears in the driver's provider list.

4.7 Platform admin: onboarding an operator

1. Signs the contract and records the commercial terms (commission or SaaS fee).
2. Configures the operator's branding, payment settlement (to the platform or to the operator's own merchant account), and notification sender name.
3. Runs a test day in sandbox mode with the operator's staff.
4. Go-live checklist sign-off.

4.8 Driver: gated car park with boom gates (no ticket)

1. **Entry:** the car stops at the entry boom. The ANPR camera reads the plate (or, in phase 2, the RFID sticker). The gate checks the plate in under 1 second, opens, and logs entry time, lane and photo.
2. **Settle-in buffer:** billable time starts after a short buffer set by the operator (e.g. 3 minutes) to cover finding a bay. Any free period (e.g. 30 minutes free) counts from entry.
3. **Confirm:** the driver gets a push notification: *"LT1234 entered City Mall car park at 10:02. Is this you?"* **Yes** links the visit to their account. **Not me** flags a possible copied plate and alerts the operator.

4. **While parked:** the dashboard shows a live timer, free time remaining and the running fee.
5. **Leaving:** the driver taps **End parking & pay** in the app if the stay went past the free period, or pays automatically with a saved card if they opted in. Within the free period there's nothing to pay.
6. **Exit:** the exit ANPR camera reads the plate, sees it's paid, free or a permit holder, and opens the boom. The visit closes. Duration and total fee go to the driver's dashboard and receipts.
7. **Visitors with no account:** the exit screen shows the plate, amount due and a QR code to pay by M-PAiSA, MyCash or card on their phone, with no sign-up needed. The boom opens on payment. An attendant or help button covers anyone who can't pay by phone.

5. UX design

5.1 Principles

- **Three taps to pay** for a returning driver; **one tap** from a reminder to extend.
- **Readable in full sun:** high-contrast theme, minimum 16 px body text, status shown by colour **and** icon and word (never colour alone).
- **Plain-language money and time:** always show the exact end time (*"Paid until 2:45 pm"*), not just a countdown, and the total before paying.
- **Honest offline states:** "Pending" is never shown as "Paid"; data age is always visible (*"as of 2:41 pm"*).
- **Low data:** app under 30 MB, offline map tiles for zones only, images compressed before upload.
- **Fairness by design:** before any notice, the driver sees the evidence and a clear route to dispute. Officers can't issue a notice without the minimum evidence.
- **Operator branding:** car park name and logo on the driver's session and notice screens; one consistent platform look everywhere else.

5.2 Languages and accessibility

- **Languages:** English, iTaukei and Fiji Hindi at launch for all driver-facing text, SMS and notices. Officer and admin apps in English first.
- **Accessibility:** WCAG 2.1 AA for web; screen-reader labels, dynamic text size and 48 px touch targets on mobile.
- Mobility-permit holders get exemptions applied automatically when their permit is linked to the plate.

5.3 Screen inventory

App	Screens
Driver	Welcome/sign-in · Identity check (licence, selfie) · Vehicles · Find parking (provider → zone, map, scan zone QR) · Duration & price · Checkout · Active session (countdown, pass QR, extend, end) · Notices list · Notice detail · Dispute form · History & receipts · Payment methods · Settings & language · Help
Officer	Login & device check · Shift start · Patrol map · Scan (camera, plate confirm, QR) · Status card · Book Now (evidence, offence) · Delayed Booking · Revisit list · Notices issued · Sync status · Shift summary · Agent mode (driver sign-up)
Operator admin	Dashboard · Car parks & zones (map editor) · Rate builder · Enforcement policy builder · Permits & exemptions · Staff & devices · Notices · Disputes queue · Refunds & waivers · Reports & settlement · Signage kit · Settings
Platform admin	Operators · Contracts & fees · Identity review queue · Fraud signals · Payments & reconciliation · Settlements · Users & vehicles lookup · Offence-code library · Notification templates · Feature flags · Audit log · System health

5.4 Notification catalogue

Event	Channels	Timing	Example message
Session paid	Push, in-app	Immediate	Paid until 2:45 pm, Zone SV-104. Receipt in app.
Expiry reminder	Local + push; SMS if no app	15 and 5 min before (operator-set)	Parking ends at 2:45 pm. Tap to extend.
Session expired	Push + SMS	At expiry	Your parking at SV-104 has expired. Extend now to avoid a fine.
Delayed booking started	Push + SMS (high priority)	Immediate	Top up within 10 min to avoid a FJ$20 fine.

Event	Channels	Timing	Example message
Notice issued	Push + SMS + email	Immediate	Notice PN-004512, FJ$20. Pay or dispute within 28 days.
Pay-after-booking still open	Push	Immediate after Book Now	Pay parking within 10 min and this notice will be cancelled.
Dispute decided	Push + SMS + email	On decision	Your dispute for PN-004512 was upheld. No payment due.
Notice escalating	Push + SMS	7 days before escalation	PN-004512 rises to FJ$30 on 12 Dec.

SMS is capped per user per day to control cost; transactional SMS always sends. Notices and payment receipts also go to email if the driver has one on file.

Part B – Software (build first)

6. Driver app

Flutter app for Android and iOS, plus a web app (PWA) for QR-at-signage entry and drivers who don't install apps. USSD/SMS channel for feature phones.

Feature	MVP (phase 1)	Later
Sign-in	Phone number + SMS one-time code; device binding	Email/Google/Apple sign-in
Identity check	Licence front/back + live selfie with liveness, matched by a KYC vendor; LTA registration photo per vehicle	Re-verification on risk signals
Vehicles	Add by plate; several per account; share with family drivers	Fleet accounts with cost centres
Find parking	Provider dropdown → zone; zone QR at signage; GPS suggestion	Live availability where gate data exists
Pay	Duration picker, live price, M-PAiSA, MyCash, card; receipt with VAT	Saved favourites, auto-extend up to a cap

Feature	MVP (phase 1)	Later
Session	Countdown + end time, signed QR pass, extend, end early	Widget on the home screen
Reminders	Local notifications scheduled at payment + push + SMS fallback	Smartwatch
Notices	List, evidence photos, pay, dispute with upload	Payment plans for large balances
Permits	View linked permits and exemptions	Buy monthly/season passes in app
History	Sessions, payments, notices, monthly statement PDF	Export for business expense claims
Support	Help centre, in-app chat, call	WhatsApp/Viber bot
Stickers (phase 2)	—	Buy, bind, replace RFID sticker (Part C)

Rules the app enforces: a session starts at payment confirmation, never before; one active session per plate per zone; extending keeps the same session ID; prices are calculated on the server, never trusted from the device.

7. Officer app

Flutter app for Android (iOS optional), locked to registered devices. Works fully offline (Section 14).

7.1 Identify a car

Method	Phase	How
Plate recognition (default)	1	Camera finds the plate and reads characters on the device in under 1 s; result checked against Fiji plate formats; officer taps to confirm or corrects
Type the plate	1	Manual entry with format hints
Driver's QR pass	1	Scan the pass on the driver's phone; signature checked offline
Continuous scan	1	Hold the phone up while walking a row; every plate read is listed with its status

Method	Phase	How
NFC tap on sticker	2	Tap the windscreen; cryptographic check
UHF RFID single / row sweep	2	With a UHF sled or handheld (Part C)

7.2 Plate recognition (ANPR) design

- **v1 (pilot):** Google ML Kit on-device text recognition, plus a plate-region detector and Fiji plate-format rules to correct common confusions (O/0, I/1, B/8). Free; quickest to ship.
- **v2 (after the pilot):** our own model: a YOLO-style plate detector + a character recogniser, trained on Fiji plates and exported to TensorFlow Lite. This is proprietary and not tied to a vendor. Target: 97%+ correct first read in daylight.
- **Training data:** 2,000+ plate photos collected in phase 0, growing from pilot captures (with consent and blurred surroundings) and labelled through an internal tool.
- **Confidence handling:** above the confidence threshold → pre-filled, one-tap confirm; below → the officer must check each character; a notice can never be issued from an unconfirmed read.
- **Night and glare:** auto-torch, exposure lock on the plate area, and a prompt to move closer.
- **Evidence:** the recognition frame (car + plate + time + GPS) is kept as photo 1 of any notice. Frames from cars with valid sessions are discarded within 24 hours.

7.3 Status card and actions

- **Statuses:** VALID · EXPIRING · EXPIRED (minutes overdue) · GRACE (delayed booking running) · NO SESSION · PERMIT/EXEMPT · UNREGISTERED PLATE · FLAGGED (e.g. plate seen elsewhere at the same time).
- **Shows:** plate, make/colour, vehicle photo, zone, paid until, previous notices in 90 days, data age when offline.
- **Actions:** Book Now · Delayed Booking · Warning (if the operator allows it) · Mark as "car gone" on revisit.
- **Evidence rules:** minimum 2 photos (plate + context), offence code from the operator's list, optional note; GPS, time and device are recorded automatically.
- **Optional Bluetooth printer** for councils whose by-laws need a paper notice on the car.

7.4 Patrol tools

- Beat map with zones, plus a heat map of sessions expiring soon.
- Revisit list sorted by expiry time and walking distance, with a vibrate alert.

- Shift summary; sync status bar; low-battery and stale-data warnings.
- **Agent mode** (same app, separate role): sign up drivers without smartphones, capture licence and photo, take cash-in through mobile money agent rules.

8. Operator admin (car park owner)

Web app for councils and private owners. Each operator sees only its own data.

Module	What the operator can do
Car parks & zones	Draw geofences on a map; bays, hours, public holidays, zone codes; publish/unpublish
Rate builder	Per hour or part-hour; minimum/maximum stay; time-of-day and weekday bands; free first X minutes; daily cap; vehicle classes (car, taxi, van, motorcycle); preview of the price for any duration
Enforcement policy builder	All Section 10 parameters, versioned; changes take effect at a scheduled time and never affect sessions already paid
Offence codes & penalties	From the platform library or the council's by-laws; amount, escalation steps, evidence required
Permits & exemptions	Staff, resident, mobility and season permits linked to plates, with validity dates; bulk upload by CSV
Staff & devices	Invite officers; roles (officer, supervisor, finance, admin); beats; register and revoke devices
Notices & disputes	Search notices; review evidence; decide disputes with a required reason; waivers above a set amount need a second approver
Refunds	Refund sessions or notices to the original payment method, with an audit trail
Dashboards	Revenue by zone/day/hour, occupancy (from sessions), compliance rate (scans found valid ÷ total), notices issued/paid/disputed, collection rate, officer productivity
Settlement & reports	Daily settlement statement by payment method; exports (CSV, PDF); VAT summary

Module	What the operator can do
Signage kit	Generate zone signs with QR and zone code, ready to print
Settings	Branding, notification sender name, terms & conditions link, support contacts

Operator roles: *Admin* (everything) · *Supervisor* (staff, notices, disputes) · *Officer* (Officer app only) · *Finance* (reports, settlement, refunds) · *Viewer* (dashboards only).

8.1 Passes & permits

Passes let operators decide which plates get in free, at a discount or on account, and when. The same rules apply at boom gates and in open car parks: an officer's status card shows PERMIT/EXEMPT.

Pass type	Example	Rules it can carry
Permanent permit	Staff, tenants, building management	Plates, sites, no expiry, free or discounted
Season pass	Monthly or yearly commuter pass	Valid dates; sold in the Driver app; auto-renew
Scheduled pass	Supplier trucks, Mon–Fri 6–10 am	Days, hours, maximum stay, maximum entries per day
One-time visitor pass	Hotel guest, contractor, meeting visitor	Link or QR sent by SMS; one stay or a date range
Validation	Shop or restaurant covers a customer's parking	A set amount or time, charged to the tenant
Exemption	Police, ambulance, fire, council vehicles, mobility permits	Always open; logged
Block list	Banned plates, unpaid debts	Deny, or allow with an alert (operator's choice)

Every pass carries:

- one or several plates, plus how many of those vehicles can be inside at once (e.g. a 5-plate fleet pass with 2 allowed inside together)
- sites, zones or lanes covered
- days and times, maximum stay, entry limits

- price: free, a discounted rate, or charged to an account
- who issued it, and when it expires

Who can issue passes: operator admins issue any pass. Tenant admins work through delegated sub-accounts with a quota (e.g. a shop can issue 20 delivery passes a month). Fleet managers can add or swap the plates on their own passes.

Gate and status decision order: block list → exemption → valid pass right now → parking credit or saved payment → pay at exit (gated) or normal session rules (open). Every decision is logged with the reason.

Phasing: simple permits and exemptions in the MVP; scheduled, visitor and validation passes and tenant sub-accounts with gated car parks.

8.2 Fleet accounts & deliveries

Fleet accounts (business customers)

- A company registers once and adds plates one at a time or by CSV upload. Each vehicle gets a class: car, van, light truck or heavy truck. Commercial rates can differ by class.
- **Billing is a monthly invoice to the business** (credit terms set per fleet). This is ordinary business credit, not stored value. Card payment and parking credit stay available for small fleets.
- **Fleet manager portal** (web): the company's vehicles, drivers, visits, time spent, costs and notices; spending limits per vehicle; monthly statements and exports for expense claims.
- Notices on fleet vehicles go to the fleet manager, who can pay them or assign them to a driver.

Deliveries at gated sites

1. **Pre-booked:** a tenant registers an expected delivery with plate (or supplier), date and time window, e.g. Tue 6–9 am. The entry boom opens automatically within the window, and the tenant is notified when the truck arrives and leaves.
2. **Loading rules:** commercial vehicles get a set free unloading time (e.g. 20 minutes). After that they pay a commercial rate, billed to the fleet account or charged to the tenant, as the operator sets.
3. **Unbooked trucks:** entry through the intercom or attendant, or a QR ticket at the entry screen. They pay at exit like any visitor.
4. **Overstays:** a truck past its window or free unloading time triggers alerts to the tenant and operator; extra time is charged.

5. **Tenant validation:** tenants can cover a delivery's or a customer's parking (e.g. first hour free with a FJ$50 purchase), charged to the tenant's account.

Data model additions: `fleets` (company, billing terms, managers), `fleetVehicles`, `passes` (type, plates, sites, schedule, limits, pricing, issuer, expiry), `tenants` (operator sub-accounts, quotas), `deliveryBookings` (tenant, plate/supplier, window, status), `validations`, `invoices`.

8.3 Analytics & reporting

The Operator admin includes a reporting module built on BigQuery and Looker Studio. Each role sees only its own data: tenants see their deliveries and validations, and fleet managers see their own vehicles.

Area	What it shows
Occupancy & traffic	Live count inside; entries/exits by hour and day; peak times; bay turnover
Time spent	Stay length by visitor, pass holder, fleet and delivery
Revenue	By site, zone, vehicle class, pass type, payment method and day part; free vs paid stays; validation costs by tenant
Fleet activity	Visits, time spent and cost per company, vehicle and driver; most frequent vehicles
Deliveries	Booked vs arrived, on-time rate, overstays, unloading time per tenant and per supplier
Passes	Active passes, how much each is used, expiring soon, unused passes
Compliance	Overstays, unpaid exits, notices, disputes, block-list hits
Gate performance	Plate read rate, manual openings, intercom calls, offline time per lane
Officers (open car parks)	Cars checked, notices issued, delayed bookings resolved, notices overturned on dispute
Parking credit	Top-ups, deductions, balances held, auto top-up use

Every view filters by date range, site and category, and can be scheduled as a weekly or monthly email, exported to CSV or PDF, or shared as a tenant-specific report.

9. Platform admin (us)

Module	What it does
Operators	Create operators; contract terms (commission %, SaaS fee, sticker revenue share); settlement model; enable/suspend
Identity review	Queue of identity checks the vendor flagged; licence vs selfie vs registration certificate; approve, reject or request more
Fraud & risk	Signals: one licence on several accounts, one plate claimed by two accounts, plate seen in two places at once, unusual dispute or refund rates, payment failures from one card
Payments	Gateway configuration per method; transaction search; nightly reconciliation exceptions; chargebacks; settlement runs and payouts
Support console	Look up user, vehicle, session or notice; reissue receipts; act on a user's behalf (logged, time-limited, reason required)
Libraries	Offence-code templates, notification templates in three languages, rate-table templates
Feature flags	Turn features on per operator (e.g. Delayed Booking, USSD, RFID) using Firebase Remote Config
Devices	Inventory of officer phones (and later RFID readers); remote lock/wipe via mobile device management (MDM)
Audit & health	Append-only audit log search; service health, error rates, payment success rate, SMS delivery rate
Stickers (phase 2)	Batch import, reseller allocation, lifecycle, replacement queue (Part C)

Platform roles: *Super admin* (two people only, hardware security key required) · *Ops* · *Support* · *Identity reviewer* · *Finance* · *Read-only auditor*.

10. Enforcement engine

Enforcement runs as two server-side state machines (session and notice). The car park owner configures every rule; the officer only picks Book Now or Delayed Booking.

10.1 Session states

From	Event	To
—	Payment confirmed	ACTIVE
ACTIVE	Time left ⩽ "expiring" threshold	EXPIRING
ACTIVE / EXPIRING	Extended (payment confirmed)	ACTIVE (same session, new end time)
ACTIVE / EXPIRING	End time passes	EXPIRED
ACTIVE	Driver ends early	ENDED (refund if the operator allows it)
EXPIRED	Officer starts Delayed Booking	GRACE
GRACE	Driver pays within window (charged from the original end time)	ACTIVE → delayed booking RESOLVED
GRACE	Window ends unpaid	EXPIRED → notice auto-issued, or put on the officer's revisit list (operator setting)

10.2 Notice states

From	Event	To
—	Officer submits Book Now with evidence	ISSUED
ISSUED	Driver pays parking + admin fee within the pay-after-booking window	CANCELLED – PARKING PAID

From	Event	To
ISSUED	Driver pays the penalty	PAID
ISSUED	Driver disputes	DISPUTED
DISPUTED	Operator upholds / rejects	CANCELLED / ISSUED (dispute clock stops while open)
ISSUED	Escalation date passes	ESCALATED (amount rises)
ESCALATED	Final date passes	REFERRED (to council debt process or private collection)
Any	Sync finds a payment confirmed before the notice time	VOID – PAID BEFORE NOTICE
Any	Waiver approved	WAIVED

10.3 Operator-configurable parameters (example defaults)

Parameter	Example default
Expiry reminders	15 and 5 min before
"Expiring" threshold for officers	10 min
Book Now enabled	Yes
Pay-after-booking window	10 min
Admin fee when paying after booking	FJ$5
Delayed Booking enabled	Yes
Delayed-booking window	10 min (choices: 5, 10, 15, 20, 30)
On delayed-booking expiry	Officer confirms (councils) / auto-issue (private)
Delayed bookings per plate per month	3, then Book Now only
Penalty	Per offence code
Payment due	28 days
Escalation	+50% after 28 days; referral after 56 days
Dispute window	14 days from issue
First-offence warning	Off

Amounts are placeholders. Councils must use the figures in their by-laws; private operators use the figures in their posted terms.

10.4 Edge cases

- **Car leaves during GRACE:** the delayed booking stays open; if unpaid at expiry and the officer marks "car gone", the operator setting decides between issuing on the evidence already taken and cancelling.
- **Two officers scan the same car:** the second sees the existing delayed booking or notice; a duplicate notice for the same plate, zone and overstay is blocked.
- **Payment arrives mid-booking:** a server timestamp decides; if payment confirmation precedes the notice submission, the notice is voided automatically.
- **Wrong plate confirmed by the officer:** caught at dispute; the photo evidence is the tie-breaker; the officer's error rate is tracked.
- **Unregistered plate:** a notice can still be issued against the plate; the operator sends it through the council process or posts it on the car (printer), as allowed.

11. Payments

Method	Integration	Notes
M-PAiSA (Vodafone Fiji)	Merchant account; M-PAiSA support issues a Client ID and Merchant Secret	No public sandbox listed, so request a test environment in phase 0; 500,000+ registered users
MyCash (Digicel Fiji)	MyCash E-Commerce REST API: `paymentRequest` → `sendOTP` → `approvePayment`	Needs a one-time code screen in the app
Cards (Visa/Mastercard, incl. M-PAiSA Mastercard)	A local bank internet payment gateway (e.g. BSP IPG) with hosted checkout and tokens	Card details never reach our servers
USSD / SMS	Via each telco's mobile money billing	Needs shortcode agreements

Design rules

- **Pay per session in phase 1.** Holding prepaid balances may count as stored value or e-money under the Reserve Bank of Fiji. Parking credit (Section 11.1) follows once the RBF confirms how it can be structured.

- **Two settlement models:** (a) we are merchant of record and pay operators daily minus commission; (b) the operator's own merchant account collects, and we bill a SaaS fee. Councils will likely want (b).
- **Idempotency:** every payment request carries an order ID; callbacks are verified (signature or secret) and processed once; retries never create two sessions.
- **Timeouts:** a payment pending more than 2 minutes shows "still checking" and keeps polling; a session only starts on confirmation.
- **Refunds** go back to the original method; partial refunds supported for early ends.
- **Nightly reconciliation:** provider statements vs our ledger; mismatches go to Platform admin with a reason code.
- **Fees:** mobile money and card fees are recorded per transaction so operator statements show gross, fees and net.

11.1 Parking credit wallet

Drivers keep a prepaid parking credit balance, topped up from M-PAiSA, MyCash or card, and parking is deducted automatically. It's how gated exits become hands-free and lets drivers budget for parking.

Regulatory position. The National Payment System Act 2021 defines electronic money as stored monetary value issued on receipt of funds for making payments. Since 30 September 2022, entities offering e-money issuance must apply to the Reserve Bank of Fiji for a payment service provider licence, and the RBF asks anyone unsure whether they need a licence or an exemption to contact it. A balance usable across many operators could count as e-money, so **a written RBF position is required before launch.** This is not legal advice.

Option	How it works	Licence risk	Effort
A. Parking credit (recommended)	Credit usable only for parking on the platform; not transferable or cashable; funds held in a bank trust account	Low–medium; may qualify as limited-purpose, so confirm with the RBF	Medium
B. Auto-pay from M-PAiSA / MyCash	Driver authorises recurring debits from their own mobile money wallet; we hold no funds	Lowest (telcos are already licensed)	Depends on providers offering pre-authorised debits

Option	How it works	Licence risk	Effort
C. Own licensed e-wallet	We become an RBF-licensed e-money issuer	We'd hold the licence ourselves	High; only if it joins a wider super-app strategy

Launch with A, worded as parking credit, never as a wallet for general use. Ask Vodafone and Digicel about B in parallel.

Features

- **Top up** in fixed amounts (FJ$10, 20, 50, 100) by M-PAiSA, MyCash or card. Larger top-ups also cut per-transaction fees on FJ$1–2 parking charges.
- **Auto top-up** from a saved card when the balance falls below a driver-set level. Mobile money top-ups stay manual, because each needs the driver's approval.
- **Automatic deductions:** on gate exit (then the boom opens); on auto-extend in open car parks (opt-in, with a cap such as 2 extra hours). Notices are paid only when the driver taps Pay.
- **Budgeting:** monthly spending limit, low-balance alerts, monthly statement, spending by car park and vehicle.
- **Shared balances:** family and fleet balances shared across several vehicles, with each driver's spending shown.
- **Safeguards:** no negative balance, except a small set gate overdraft so nobody is stuck at a barrier; PIN or biometric for top-ups; full history; remaining balance refundable when an account closes.
- **Ledger:** every top-up and deduction is a separate entry, never just an edited balance. It's reconciled every night against the trust account and the payment providers.

Phasing: RBF enquiry in phase 0; build after a positive response, targeted for the commercial launch (phase 6) or gated car parks, whichever comes first. Data model additions: `creditAccounts`, `ledgerEntries` (append-only), `topUps`, `autoTopUpRules`.

12. Architecture & tech stack

One multi-tenant backend on Google Cloud / Firebase, hosted in **australia-southeast1 (Sydney)**, the closest Google Cloud region to Fiji. Clients talk to Firestore directly for reads (guarded by security rules) and to Cloud Functions for anything involving money, enforcement or identity.

Layer	Choice	Why
Driver app	Flutter (Android, iOS) + Flutter web PWA	One codebase; strong offline support
Officer app	Flutter (Android first) with native plugins for camera ANPR and, in phase 2, the Chainway UHF SDK	Same codebase skills; native speed where needed
Admin web (Operator + Platform)	Next.js (React) on Firebase App Hosting	Rich tables, maps and dashboards
Auth	Firebase Auth (phone one-time code for drivers; email + multi-factor for staff; hardware key for super admins); custom claims for roles	Managed, proven
App integrity	Firebase App Check (Play Integrity / App Attest)	Blocks scripted or modified clients
Data	Cloud Firestore (operational), Cloud Storage (photos, identity media in a separate locked bucket)	Real-time sync + offline cache built in
Business logic	Cloud Functions (2nd gen, TypeScript)	Payments, sessions, notices, disputes, settlement
Timers	Cloud Tasks (one task per reminder / grace window)	Exact timing without polling
Partner API & heavy jobs	Cloud Run (REST API, server-side ANPR fallback, PDF generation)	Scales to zero; containerised
Messaging	Firebase Cloud Messaging; SMS via two local gateways (primary + backup); email via a transactional provider	No single point of failure for alerts
Analytics	Firestore → BigQuery export; Looker Studio dashboards; BigQuery ML later for demand forecasting	Cheap, scalable reporting

Layer	Choice	Why
Config	Firebase Remote Config (feature flags per operator)	Turn on RFID, USSD, Delayed Booking per operator
Secrets & keys	Secret Manager; Cloud KMS for signing keys and customer-managed encryption	Keys never in code
Observability	Cloud Logging, Monitoring, Error Reporting, Crashlytics, Performance Monitoring	One place to see health
CI/CD	GitHub + Cloud Build; separate dev, staging, production projects	Safe releases

Key flows

- **Pay:** app → `createCheckout` function (server prices the session) → provider → callback → `confirmPayment` (idempotent) → session ACTIVE → Cloud Tasks reminders scheduled → signed QR pass issued.
- **Check a car:** officer app reads plate → local cache lookup (instant) → live Firestore lookup if online → status card.
- **Book:** officer app writes a notice draft with evidence to a queue → `issueNotice` validates evidence, policy version and duplicate rules → notice ISSUED → notifications.

13. Data model & APIs

13.1 Firestore collections

Collection	Key fields
`operators`	name, type (council/private), branding, settlement model, fees, status
`carParks`, `zones`	operatorId, geofence, bays, hours, zoneCode, rateTableId, policyId
`rateTables` (versioned)	bands[], unit, min/max stay, daily cap, class overrides, effectiveFrom
`policies` (versioned)	all Section 10 parameters, offence codes and amounts, effectiveFrom

Collection	Key fields
`permits`	operatorId, plate, type, validFrom/To
`users`	phone, name, language, identityStatus, identityRef (no images in Firestore)
`vehicles`	plate (normalised, unique per country), make, model, colour, ownerUserId, verified, tagIds[] (phase 2)
`sessions`	plate, vehicleId, operatorId, zoneId, start, end, amount, paymentIds[], status, policyVersion
`payments`	orderId (idempotency key), provider, providerRef, amount, fees, status, settlementId
`delayedBookings`	plate, sessionId, officerId, createdAt, expiresAt, status, evidence[]
`notices`	noticeNo, plate, offenceCode, amount, status, officerId, deviceId, gps, photos[], policyVersion, timeline[]
`disputes`	noticeId, reason, attachments, decision, decidedBy, decidedAt
`staff`, `devices`	operatorId, roles, beats, deviceId, integrity status, lastSync
`scans`	officerId, method, plateRead, confidence, confirmedPlate, statusShown, gps, time
`settlements`	operatorId, period, gross, fees, net, payout ref
`auditLog`	actor, action, target, before/after, time, reason (write-only from functions)
`tags` (phase 2)	tid, epc, nfcUid, shortCode, batchId, status, vehicleId

13.2 Partner REST API (Cloud Run, phase 3)

Endpoint	Purpose
`GET /v1/status?plate=&zone=`	Parking status for a plate (gate systems, council tools)

Endpoint	Purpose
`POST /v1/gate-events`	Entry/exit events from barriers, fixed ANPR or RFID readers
`POST /v1/sessions`	Start a session from a partner channel (e.g. a kiosk)
`GET /v1/settlements`	Operator settlement data for their finance systems
Webhooks: `session.started`, `session.expired`, `notice.issued`, `notice.paid`, `payment.settled`	Push events to partner systems

Auth: OAuth 2.0 client credentials per partner, scoped to one operator; rate-limited; every call is logged.

14. Offline mode & sync

Every app is offline-first: it reads from a local database (Firestore offline persistence, plus SQLite for the officer's zone snapshot), queues writes, and syncs when connected. Basements, multi-storey car parks and areas outside town centres must work.

App	Works offline	Needs a connection
Driver	Vehicles, rates, zones, the active session and countdown, signed QR pass, notices and receipts already downloaded; local expiry reminders	Paying or extending (shown as Pending until confirmed); USSD/SMS works with signal but no data
Officer	Plate recognition, zone snapshot of sessions/permits/flags, QR pass verification, Book Now and Delayed Booking with evidence (queued), revisit list timers	Live status for cars paid after the last snapshot; uploading notices
Operator / Platform admin	Cached dashboards and lists (read-only)	Any change to rates, policies, staff, disputes

Officer snapshot: active sessions, permits and flagged plates for assigned zones, refreshed every 60–120 s while online (changes only), plus the public key for checking QR passes. The status card always shows *"offline – data as of HH:MM"* when relevant. Snapshots older than a limit set by the operator (e.g. 15 min) block Book Now without a second confirmation.

Signed QR pass: plate, zone, start, end and session ID, signed with Ed25519 by Cloud KMS. The officer app verifies it with no network. Passes are valid only until the session end time.

Sync & conflict rules

1. Queued officer records carry a device-signed timestamp, GPS and the policy version used.
2. On upload, the server re-checks each notice: if a confirmed payment covers the time of the notice, the notice becomes VOID – PAID BEFORE NOTICE and both parties are told.
3. Delayed-booking timers run on the device and the server; **the server clock decides**.
4. Duplicate notices (same plate, zone, overstay) from two offline officers: the earliest is kept, the other is voided.
5. Offline records older than 24 hours are flagged for supervisor review before they become notices.
6. Admin changes made while offline are queued and applied only after the server accepts them, with the new policy version.

15. Security

The system handles money, identity documents and legal notices, so it's built to OWASP MASVS (mobile) and ASVS Level 2 (web/API) and gets an independent penetration test before the pilot.

15.1 Threat model

Threat	Impact	Controls
Driver account takeover (SIM swap, stolen phone)	Fines routed wrongly, payment misuse	Device binding; re-verify on new device; step-up check for plate or payment-method changes; alerts on login from a new device
Someone registers another person's plate	Victim gets fines; offender avoids them	LTA registration photo + identity check before notices route to an account; one verified owner per plate; dispute override

Threat	Impact	Controls
Fake or copied plates	Wrong person fined	Officer photo evidence; "seen in two places" flag; phase 2 RFID identity
Stolen officer phone	Fake notices, data exposure	Device registration, PIN/biometric, short sessions, remote lock/wipe (MDM), local data encrypted, App Check
Officer abuse (fake or revenge notices)	Unfair fines, reputation	Mandatory evidence; GPS/time stamped by the device and checked against the server; officer error and dispute rates tracked; supervisor review
Insider waiver/refund fraud	Revenue loss	Role limits; two-person approval above a threshold; every action in the append-only audit log; weekly exception report
Payment fraud, replayed callbacks	Free parking, double sessions	Server-side pricing; verified callbacks; idempotency keys; amount and order checks
Forged QR parking pass	Free parking	Ed25519 signatures from Cloud KMS; pass bound to plate and end time; officer app verifies
Identity data breach	Serious harm to drivers	Separate bucket, customer-managed keys, no public access, signed short-lived URLs only for reviewers, access logged, retention limits
API abuse, scraping, denial of service	Outage, data leak	App Check, rate limits, Cloud Armor on public endpoints, partner OAuth scopes
Tenant data leakage between operators	Contract and privacy breach	operatorId on every record; Firestore rules tested automatically; no cross-tenant queries from clients

15.2 Access control (summary)

Capability	Driver	Officer	Op. supervisor	Op. admin	Op. finance	Support	Platform super admin
Pay / view own sessions	Yes	—	—	—	—	View	View
Check a car's status	Own only	Own operator	Own operator	Own operator	—	View	View
Issue notices	—	Yes	Yes	Yes	—	—	—
Decide disputes / waive	—	—	Yes (limit)	Yes	—	—	—
Refunds	—	—	—	Yes	Yes	Request	Yes
Rates & policies	—	—	—	Yes	—	—	—
Identity documents	Own	—	—	—	—	Reviewer role only	Reviewer role only
Operators & fees	—	—	—	—	—	—	Yes

15.3 Encryption & keys

- TLS 1.2+ everywhere; certificate pinning in mobile apps for our API domains.
- Google encryption at rest by default; **customer-managed keys (Cloud KMS)** for identity media and signing keys; key rotation every 12 months.
- Secrets in Secret Manager only; no keys in apps or code; provider credentials scoped per environment.
- On devices: encrypted local storage (Android Keystore / iOS Keychain); queued evidence encrypted until uploaded.

15.4 Card data

Card payments use the bank gateway's hosted checkout and tokens, so card numbers never touch our systems. This keeps us at the lightest PCI DSS self-assessment level (SAQ A).

15.5 Security operations

- Staff accounts: multi-factor required; super admins use hardware security keys; access reviewed quarterly.
- Dependency and container scanning in CI; mobile release builds obfuscated.

- Penetration test before the pilot and yearly after; a responsible-disclosure email address.
- Incident response plan: severity levels, on-call contact, user and operator notification within 72 hours of a confirmed personal-data breach.

16. Reliability, performance & scaling

16.1 Service targets

Measure	Target
Core API availability (pay, status, notices)	99.9% per month (about 43 min downtime max)
Status lookup, server side	p95 under 300 ms
Camera to status card (online, incl. confirm)	under 1.5 s
Payment confirmation	p95 under 10 s (depends on provider)
Reminder delivery	99% within 60 s of schedule
Driver app cold start (mid-range Android)	under 3 s
Officer app offline capability	a full shift with no signal

16.2 Failure handling

Failure	What users see	Fallback
One payment provider down	That option greyed out with a notice	Other methods; USSD; operator may apply a short grace window automatically
All payments down	"Payments unavailable since 2:10 pm" banner	Automatic platform-wide grace: no notices for unpaid time during the outage (logged)
SMS gateway down	Nothing	Automatic switch to the backup gateway; push still works
Push service down	Nothing	SMS for time-critical alerts

Failure	What users see	Fallback
Mobile data down for officers	Offline banner with data age	Offline mode (Section 14)
Cloud region problem	Short outage	Restore in the second Australian region from backups

16.3 Backups & recovery

- Firestore point-in-time recovery (7 days) plus daily scheduled backups kept for 90 days; Cloud Storage versioning for evidence photos.
- **Recovery point objective:** at most 5 minutes of data lost. **Recovery time objective:** 4 hours. Restore drill every quarter.
- Infrastructure as code (Terraform), so environments can be rebuilt.

16.4 Monitoring & on-call

- Dashboards and alerts for: error rate, payment success rate per provider, callback delays, SMS delivery rate, task queue backlog, app crash-free rate (target 99.5%+).
- Uptime checks on public endpoints; status page for operators.
- On-call rota with a 30-minute response for P1 incidents during operating hours (6 am–10 pm Fiji time) and post-incident reviews.

16.5 Scaling

National scale is small for this stack: even 50,000 paid sessions a day is under one write per second on average. Design for peaks (8–9 am, lunch) with sharded counters for dashboards, no hot documents, and Cloud Tasks for timers. Costs grow with usage, so the pilot stays inside or near the free tier.

17. Privacy, legal & compliance

- **Legal basis for council notices:** by-laws under the Local Government Act 1972, which is under review. Each council must confirm that digital notices, electronic service and Delayed Booking are valid, or amend its by-laws. Suva itself is waiting on law amendments before digital enforcement of inner-city streets.
- **Private car parks** can't issue statutory fines. They charge a contractual "parking charge" under terms on entrance signage and accepted in the app. We supply a template for legal review.

- **Personal data:** collect only what is needed; explicit consent screens for identity checks and photos; drivers can download and delete their data (subject to open notices and legal retention). Get legal advice on current Fiji privacy obligations and build to a GDPR-equivalent standard regardless.
- **Retention schedule (proposed):** scan frames without a notice 24 hours; notice evidence 2 years after closure; identity images 30 days after verification (keep the result, not the images, unless the law requires more); financial records 7 years.
- **Stored value:** no prepaid wallet until the Reserve Bank of Fiji position is confirmed.
- **Radio licence (phase 2):** UHF RFID readers need TAF approval (Part C).
- **Terms & policies to draft:** driver terms, privacy policy, operator agreement, officer code of conduct, data processing terms with each operator, dispute and appeal procedure.

18. Operations & support

18.1 Customer support

Channel	Hours	First response target
In-app help centre (EN, iTaukei, Fiji Hindi)	24/7	Self-service
In-app chat / WhatsApp / Viber	7 am–7 pm, Mon–Sat	15 min
Phone line	8 am–5 pm, Mon–Fri	5 min wait
Email	Any time	1 working day
Agents and council counters	Their hours	In person

Top support cases to script: payment taken but no session, wrong plate entered, notice received while paid, can't receive the one-time code, lost phone, car sold.

18.2 Disputes & appeals

1. Driver disputes within 14 days with a reason and evidence; the notice's payment clock stops.
2. Operator supervisor decides within 10 working days, with a written reason.
3. If rejected, the driver has 14 days to pay the original amount (no escalation during the dispute).
4. Councils: further appeal follows the council's legal process; private operators: an independent review option is recommended for credibility.

18.3 Operator onboarding & training

- Operator admin training: half a day, remote or on site.
- Officer training: half a day plus a supervised first shift; a pocket guide covering evidence rules, conduct and offline mode.
- **Signage kit:** zone signs with QR, zone code, rates and terms link; entrance terms sign for private car parks.
- Launch support: two weeks of daily check-ins, then a weekly report.

18.4 Driver adoption

- Launch campaigns with the operator (mall screens, radio, social media); first-session promo credit funded by the operator or by us.
- Agents help drivers without smartphones sign up.
- Grace period at launch: warnings instead of notices for the first 2 weeks at each new car park.

19. Testing & quality

Test type	What	When
Unit & integration	Pricing engine, state machines, policy versions, idempotent payments (Firebase Emulator Suite)	Every commit
Security rules tests	Tenant isolation, role permissions	Every commit
End-to-end	Driver pay → officer check → notice → dispute, on real devices	Every release
ANPR accuracy	Labelled Fiji plate test set; first-read accuracy by day/night/rain	Every model change
Offline drills	Officer full shift in airplane mode; sync conflict scenarios	Each release before the pilot
Payment sandbox & live tests	Every provider: success, failure, timeout, refund, duplicate callback	Phase 4
Load test	10× expected peak sessions and scans	Before pilot and before each new council
Penetration test	Independent firm, apps + APIs + admin	Before pilot, then yearly

Test type	What	When
User acceptance	Operator staff and 10–20 drivers try every journey	End of phase 4
Accessibility & language review	Screen reader, contrast, translations checked by native speakers	Before pilot

Pilot measurement plan: first-read plate accuracy ≥ 95% (100% after officer confirmation), camera-to-status ≤ 1.5 s, payment success ≥ 95%, driver sign-up to first payment ≤ 2 min, dispute rate < 5% of notices, officer time per car at least halved versus the booking book, crash-free sessions ≥ 99.5%.

Part C – RFID hardware (phase 2 upgrade)

20. Why add RFID, and for whom

RFID is an optional premium layer on top of the plate system: identity stays with the plate, and the sticker is a faster, tamper-proof key to the same vehicle record.

- **For drivers:** hands-free entry at gated car parks, fewer wrongful notices from misread plates, a "verified vehicle" badge. Likely buyers: fleets, taxis, rental companies, daily commuters.
- **For operators:** row-sweep enforcement (a whole row in seconds), works when plates are dirty or blocked, makes copied plates obvious.

21. The windscreen sticker

Passive UHF windscreen tags are a commodity product: EPC Gen2 / ISO 18000-6C, 860–960 MHz, no battery, tamper-proof versions that stop working when peeled off. Vendors quote up to 8–10 m read range with fixed readers; one vendor measured about 21 ft (~6 m) with a full-power handheld.

Item	Specification
UHF protocol	EPC Class 1 Gen2 / ISO 18000-6C
UHF chip	Impinj Monza R6/M7xx or NXP UCODE 8; the factory-locked **TID** is the identity
Tuning	Optimised for **915–925 MHz** (Fiji allocation)

Item	Specification
NFC inlay	NXP **NTAG 424 DNA**: one-time cryptographic signature per tap, can't be cloned
Tamper	Frangible; both inlays break on removal
Printed face	Brand, QR deep link (signed), 8-character short code, "Non-transferable", UV laminate
Size	~70×40 mm to 85×54 mm
Mounting	Inside the windscreen behind the mirror; headlamp or plate-mount variant for metallised windscreens and metal tint
Retail pack	Sticker + card with a scratch-off one-time activation PIN

First test: samples from 3–4 suppliers, read-tested through windscreens common in Fiji (Japanese imports, tinted glass, vans and buses with heat-reflective glass).

22. Binding, lifecycle & replacement

22.1 Binding methods (no NFC needed)

Method	Device	How it works
NFC tap	Smartphone with NFC	Tap the sticker; cryptographic check; strongest
QR scan + PIN	Any camera smartphone	Scan the sticker's signed QR, enter the scratch PIN
Short code + PIN	Any browser	Type the 8-character code and PIN on the website
USSD / SMS	Any mobile phone	`BIND <code> <PIN> <plate>` to a shortcode
Assisted	None (customer)	Agent, council counter or partner windscreen shop reads the sticker and binds it

After binding: an install photo with the plate visible goes to the verification queue. The sticker locks to the first successful binding. Because the plate is already verified, no new identity check is needed.

22.2 Lifecycle

Unassigned → Allocated (to a reseller) → Bound → Active → Replaced / Lost-Stolen / Destroyed. Officers who scan a Replaced or Lost sticker see *"Replaced tag – check plate"*, not a fraud alarm.

22.3 Replacement (customer experience)

1. Report in the app (*Vehicle → Sticker → Replace*), by USSD, at an agent or by phone; choose a reason.
2. The old sticker deactivates instantly; **parking continues by plate**, with a banner saying so.
3. Collect a new sticker from a sales point, have it couriered, or have it fitted by a partner windscreen shop.
4. Bind with any method above, then take a new install photo. Done in about 2 minutes.

Situation	Suggested price
Windscreen replaced	Discounted, or free through partner shops
Faulty or faded	Free within 12-month warranty
Tint or metallised glass blocks reads	Free swap to headlamp/plate-mount version
Lost or vandalised	Full price
Car sold	New owner buys their own; old sticker destroyed on removal

More than 2 replacements per vehicle in 12 months triggers manual review.

23. Readers & gated car parks

Hardware	Use	Price guide
Chainway R6 UHF sled (clips onto the officer's phone)	Low-cost RFID enforcement	Roughly €310–600
Chainway C72 rugged Android handheld with Impinj E710 UHF (865–928 MHz), IP65, 8000 mAh	Full-shift council enforcement	Roughly US$800–1,200
Fixed UHF reader + antenna at a barrier	Hands-free entry/exit	US$600–1,500 per lane (estimate)

Hardware	Use	Price guide
Fixed ANPR camera at a barrier	Ticketless entry/exit by plate	Quote-based

Row-sweep design: the app filters reads by signal strength (RSSI), shows plate and vehicle photo for confirmation, and lists every sticker in range with its status colour.

23.1 Boom gate system for private car parks

Private gated car parks run on **entry–exit billing**: the clock starts when the boom reads the plate, and the barrier at the exit guarantees payment. It's plate-based, so it **doesn't need RFID** and can launch right after the phase 1 software. RFID gate readers are an optional add-on in phase 2.

Hardware per lane

Component	Purpose	Price guide (USD, estimate)
Motorised boom barrier (3–6 m arm), with safety photo-eye, manual release and battery backup (UPS)	Physical control of entry and exit	$800–3,000
ANPR camera with infrared lighting, set for the lane	Reads plates day and night at 0–15 km/h	$300–1,000
Vehicle detector (ground loop or radar)	Detects a car is present; stops the boom closing on it	$100–400
Gate controller (industrial mini PC in an IP65 enclosure, 4G + wired network)	Runs our gate software; makes the open/stay-closed decision locally	$300–800
Exit display (LCD/LED) with pay QR	Shows plate, time, amount and how to pay	$200–800
Help/intercom button	Calls the attendant or our support line	$100–300
Optional: UHF RFID reader + antenna (phase 2)	Hands-free entry for sticker holders	$600–1,500

Installation, civil works (islands, cabling, loops) and signage are extra and site-specific. Get local installer quotes.

Gate software (our own component)

- A **gate agent** runs on the controller at each site. It reads camera events, decides in under 1 s, and triggers the barrier relay. It's built and owned by us, not a camera vendor's cloud.
- It keeps a **local cache** of permits, active visits, paid exits and blocked plates, so gates keep working when the internet drops. Events queue and sync on reconnect.
- It sends events to the platform (`POST /v1/gate-events`, or Pub/Sub) and receives "paid" signals in real time.
- Each operator chooses the offline policy: open for known plates and permits only, or open for all and charge on reconnect.

Operator settings (Operator admin)

Setting	Example
Free period	30 min from entry
Settle-in buffer before billing starts	3 min
Exit grace after paying (time to reach the exit)	15 min, then extra time is charged
Rates and daily maximum	FJ$1.50/h after free period, FJ$12 max per day
Permits and whitelists	Staff, tenants, VIPs open automatically
Auto-charge on exit	Allowed for opted-in drivers with a saved card
Plates with unpaid notices	Allow entry with a reminder, or deny (operator's choice)
Hours and after-hours behaviour	Locked, exit-only, or open

Edge cases

- **Plate not read at entry** (dirty, damaged, motorcycle): the entry screen shows a QR "ticket" to scan with a phone, or the attendant enters the plate. At exit, any visit without a read is matched by entry photo and time.
- **Misread at exit:** fuzzy-match against cars currently inside (e.g. one character off) plus the photo. If it's still unclear, the attendant confirms.
- **Tailgating:** the vehicle detector counts cars. A car exiting without a matching entry triggers an alert and a payment prompt.
- **Paid but exit grace ran out:** the extra minutes show on the exit screen, payable by QR.

- **Power failure:** the UPS keeps barriers and controller running. On a long outage, barriers are raised manually (fail-safe for emergencies) and the operator is alerted.
- **Driver says "Not me":** the visit is held for review, and the account owner isn't charged until it's resolved.

Data model additions: `sites`, `lanes` (entry/exit, devices), `gateEvents` (plate, confidence, photo, lane, time), and visits as a session type with `entryAt`, `billableFrom`, `exitAt`, `paidUntil`, `exitGraceUntil`.

24. Radio spectrum & import

The APT regional RFID table lists Fiji's UHF RFID allocation as **915–925 MHz at 1 W EIRP, with registration and an individual licence required**. That table may be out of date, so confirm with the Telecommunications Authority of Fiji (TAF) in phase 7a. TAF's fee schedule includes an apparatus approval fee. Readers must be set to the Fiji band and power before import and deployment.

Part D – Delivery

25. Project timeline (software first, then RFID)

Software reaches a pilot-ready MVP in week 16, the plate-only pilot runs weeks 16–24, and RFID hardware moves from evaluation (from week 20, in parallel) to its own pilot by week 42. Week 1 = project start; each phase ends at a gate before the next commitment.

Phase	Weeks	Scope	Exit criteria / gate
0. Mobilise & discovery	1–3	Confirm pilot site and operator; requirements and wireframes for all four apps; Firebase dev/staging/production projects, CI/CD; apply for M-PAiSA, MyCash and card gateway merchant/test accounts; USSD/SMS shortcode enquiries; KYC vendor shortlist; start collecting 2,000+ Fiji plate photos	Signed pilot letter of intent; wireframes approved; merchant applications lodged

Phase	Weeks	Scope	Exit criteria / gate
1. Core platform + Driver MVP	3–8	Data model, security rules, pricing engine, session state machine, Cloud Tasks reminders, signed QR pass; Driver app and web app: sign-in, vehicles, zones, pay (card sandbox), session, reminders, history; identity check integration	A driver can pay and receive reminders end-to-end in staging
2. Officer app + ANPR v1	6–11	Officer app: login, device registration, patrol map, ANPR (ML Kit), plate confirm, QR pass check, status card, Book Now, Delayed Booking, evidence, revisit list, offline snapshot and queue; notice state machine	Officer completes a full simulated shift offline; ANPR ≥ 90% first read on the test set
3. Operator & Platform admin	9–13	Zones map editor, rate builder, policy builder, staff/devices, notices, disputes, refunds, dashboards, settlement reports, signage kit; Platform admin: operators, identity queue, payments, support console, audit	Operator can set up a car park in under a day without our help
4. Payments live, hardening & UAT	12–16	Live M-PAiSA, MyCash and card; reconciliation; SMS gateways; translations (EN/iTaukei/Fiji Hindi); load test; penetration test and fixes; accessibility review; UAT with operator staff and 10–20 drivers; terms and privacy policy	**Gate G1 – Pilot ready:** zero open critical/high security findings; all journeys pass UAT; payment success ≥ 95% in live tests
5. Plate-only pilot	16–24	One private car park, 50–100+ registered vehicles; 2 weeks of warnings, then live enforcement; weekly KPI reviews; ANPR v2 training on pilot data	**Gate G2 – Go/no-go:** pilot KPIs met (Section 19); operator willing to sign a commercial contract

Phase	Weeks	Scope	Exit criteria / gate
6. Commercial launch v1	24–32	Fix pilot findings; ANPR v2 (own model); onboard 2–3 more private operators; USSD channel; operator self-service onboarding; marketing launch	3+ paying operators live; crash-free ⩾ 99.5%
7a. RFID evaluation (parallel)	20–28	Sticker samples from 3–4 suppliers; windscreen read tests; reader trials (R6, C72); TAF licence and type approval; choose supplier; design sticker artwork and retail pack	Supplier chosen; read rate ⩾ 98% on target windscreens; TAF approval in hand
7b. RFID software	28–36	Chainway SDK plugin in the Officer app; row-sweep; NFC tap; sticker inventory, binding (all methods), lifecycle and replacement flows; Platform admin sticker module	All binding and replacement journeys pass UAT
7c. RFID pilot	36–42	200–500 stickers at a pilot operator; measure enforcement speed vs plate-only; sticker sales via agents and windscreen shops	**Gate G3:** RFID halves officer time per car vs ANPR; sticker unit economics positive
8. Gated car parks & partner API	40–50	Partner REST API and webhooks; fixed UHF readers and ANPR cameras at barriers; ticketless entry/exit	First gated car park live
9. Municipal scale-up	48+	Council by-law alignment, council integrations, scan car option, second and third municipalities	First council contract

Critical path: merchant API access (phase 0 → 4) and the pilot operator's commitment. Both start in week 1.

26. Team & budget

26.1 Team (software phases 0–6)

Role	Load	Main responsibility
Product owner (Jone)	Part-time	Scope, operator relationships, pilot, decisions
Tech lead / backend (Firebase, TypeScript)	Full-time	Architecture, functions, payments, security
Flutter developer	Full-time	Driver and Officer apps
Web developer (Next.js)	Full-time from week 6	Operator and Platform admin
UI/UX designer	Part-time, weeks 1–10	Wireframes, design system, translations layout
ML engineer	Part-time, weeks 6–30	ANPR v1 tuning, v2 model
QA / test engineer	Part-time from week 8	Test plans, devices, UAT, load tests
Support & operations	Part-time from week 14	Pilot support, operator training

Phase 7 adds an RFID/hardware integrator (part-time) for reader trials, TAF paperwork and gate installs.

26.2 Hardware & running costs (USD, indicative)

Item	Phase	Estimate
2–3 test Android phones + rugged cases + power banks	0–5	$600–1,200 (estimate)
Bluetooth thermal printer (optional)	5	about $100 (estimate)
Penetration test	4	quote-based
KYC/liveness vendor	1 onward	per-check pricing, quote-based
Cloud (Firebase/Google Cloud)	1 onward	near free tier in the pilot; grows with usage

Item	Phase	Estimate
SMS	4 onward	per message, from local gateways
Sticker samples (~200)	7a	$300–600 (estimate)
Chainway R6 sled × 2 / C72 handheld × 1–2	7a	about €310–600 each / US$800–1,200 each
Fixed UHF reader or ANPR camera for a barrier demo	8	$600–1,500 per lane (estimate)

Staff costs depend on in-house vs contracted developers and are not included here.

26A. Development environment & build workflow

Local or cloud? Both, with clear roles

Where	What happens there
GitHub (cloud)	The single source of truth: all code, docs, issues, pull requests, releases
Developer machines (local)	Writing code; running everything against the **Firebase Emulator Suite** (Auth, Firestore, Functions, Storage) for a fast, free test loop; Android Studio for the mobile apps; an Apple Mac with Xcode for iOS builds (or a cloud Mac build service)
Google Cloud (cloud)	Three separate projects: `dev` (shared test data), `staging` (pilot rehearsal), `prod` (live). Nothing is built by hand in prod
CI/CD (cloud)	GitHub Actions runs tests on every pull request, deploys to staging when code merges to `main`, and deploys to production from a tagged release. Test mobile builds go to testers through Firebase App Distribution

Google AI Studio stays useful for quick clickable UI prototypes in phase 0. The production system (payments, security rules, four apps) is built in the repository.

When to move to Claude Code

From day one of phase 0. Claude Code works directly in the repository, either locally in a terminal or on the web against the GitHub repo. It scaffolds the monorepo, writes code and tests, and opens pull requests for review. The repo holds a `CLAUDE.md` with the architecture decisions, coding conventions and commands, plus this design document under `/docs`, so every session starts from the same spec.

Repository layout (one monorepo)

```
/apps/driver          Flutter: Driver app (Android, iOS, web PWA)
/apps/officer         Flutter: Officer app (Android)
/apps/admin           Next.js: Operator admin + Platform admin (role-based
areas)
/packages/ui          Shared Flutter design system and widgets
/packages/models      Shared Dart models generated from the data schema
/functions            Cloud Functions (TypeScript): pricing, sessions,
payments, notices, entitlements
/services/partner-api Cloud Run REST API and webhooks (phase 3)
/services/gate-agent  On-site gate software (gated car parks)
/infra                Terraform: projects, buckets, KMS, IAM, schedulers
/docs                 Design doc (PDF + source), ADRs, runbooks
CLAUDE.md             Instructions and conventions for Claude Code
```

Build sequence

1. **Foundations (weeks 1–2):** create the repo, monorepo scaffold, `dev` / `staging` / `prod` projects, Terraform, CI pipeline, emulators, `CLAUDE.md`, coding standards.
2. **Domain core (weeks 2–4), no UI yet:** data model and security rules; pricing engine, enforcement policy engine, session and notice state machines; entitlements. All covered by automated tests, because everything else depends on them.
3. **Backend services (weeks 3–6):** auth and roles, sessions, payments (sandbox first), Cloud Tasks reminders, signed QR pass, notifications.
4. **Minimal admin (weeks 5–7):** Platform admin can create an operator and plan; Operator admin can set up zones, rates and policy. This is needed to configure real test data.
5. **Driver app (weeks 4–9):** sign-in, vehicles, find parking, pay, session, reminders, notices.
6. **Officer app (weeks 6–11):** ANPR, status card, Book Now, Delayed Booking, evidence, offline mode.
7. **Full admin + reporting (weeks 9–13):** disputes, refunds, dashboards, settlement, modules and plans.

8. **Live payments, hardening, UAT (weeks 12–16),** then the pilot.

Gate agent, parking credit, fleets and RFID follow the phases in Section 25.

Versioning: from the first commit

- **Git from day one,** trunk-based: short-lived feature branches, pull requests into `main`, protected `main` (review + passing tests required).
- **Conventional Commits** (`feat:` , `fix:` , `chore:`) so release notes generate automatically.
- **Semantic versioning per app** (Driver 0.1.0 → 1.0.0 at commercial launch); Git tags for every release; build numbers increase with every store upload.
- **API versioned** from the start (`/v1`); breaking changes only in a new version.
- **Data changes as migration scripts** in the repo, run by CI, never edited by hand in the console.
- **Business rules versioned** in the data itself (rate tables, policies, plans), so every session and notice records which version applied.
- **Architecture Decision Records** (`/docs/adr`) for key choices, e.g. "plate-first identity" and "parking credit vs wallet".

Accounts to set up in week 1

GitHub organisation (private repo), Google Cloud organisation with billing, three Firebase projects, Apple Developer Program, Google Play Console, domain and DNS, a password manager for shared secrets, and merchant/test applications with M-PAiSA, MyCash and the card gateway bank.

27. Risks & open decisions

27.1 Risks

Risk	Likelihood	Impact	Mitigation
Mobile money API access is slow	High	High	Apply in week 1; launch with cards first if needed
ANPR misreads (dirty or custom plates, night, rain)	Medium	High	Mandatory officer confirmation; Fiji-trained model; torch; QR pass and RFID as second checks

Risk	Likelihood	Impact	Mitigation
Councils' by-laws don't yet allow digital notices or Delayed Booking	High (councils)	Medium	Start with private car parks; give councils a by-law amendment brief
Bula Parki entrenched in Suva	High	Medium	Lead with private operators and other towns; offer integration, not replacement
Someone registers another person's plate	Medium	High	Registration certificate + identity check before notices route to accounts
Fake or copied plates	Low–Medium	Medium	"Seen in two places" flag; RFID phase
Low driver adoption	Medium	High	Pay by plate with nothing to buy; signage QR; agents; launch warnings period
Identity data breach	Low	Very high	Section 15 controls; minimal retention
Stored-value regulation	Medium	Medium	No wallet until RBF position is clear
RFID: metallised windscreens, several reads, EPC cloning	Medium	Medium	Alternative mounts; RSSI + photo confirm; TID + NFC crypto
TAF licensing delays for readers	Medium	Low (phase 2 only)	Start enquiry in phase 7a, in parallel with the pilot

27.2 Open decisions

1. Which entity and brand owns the product: SLEDIEN, Vibrant, or a new brand?
2. Business model: commission per session vs SaaS fee per operator; who keeps penalty revenue; sticker price.
3. Identity check: mandatory for every driver from the second session (current design), or only above a threshold?
4. Pilot site: which private car park hosts the first 50–100 vehicles?
5. Team: in-house developers, contractors, or a mix?
6. Approach Suva City Council as an integration partner now, or after the pilot?
7. KYC/liveness vendor and card gateway bank.

8. RBF position on parking credit: limited-purpose stored value, an exemption, or a licence?
9. Fleet billing: credit terms and limits for business invoices; who carries the bad-debt risk (us or the operator)?

28. Sources

- FBC News – Suva introduces real-time monitoring for city parking (Bula Parki), Jan 2026
- FBC News – Suva digital with new cashless parking app, Jan 2026
- FBC News – Traffic and parking changes for four municipalities, Jul 2026
- ContinuousCare – M-PAiSA merchant integration (Client ID / Merchant Secret)
- FRCS – M-PAiSA payment facility, 500,000+ users
- MyCash E-Commerce API document (via Issuu)
- BSP – Digicel rolls out BSP payment gateway for MyCash
- GAO RFID – UHF Gen2 tamperproof windshield tag (read-range tests)
- JIA RFID – UHF Gen2 tamperproof windshield tag spec
- POS Market – Chainway C72 specs and pricing
- rfid.it – Chainway C72 and R6 pricing
- APT – Regional RFID spectrum recommendation (Fiji 915–925 MHz)
- TAF – Licence fee schedule